Your Undergraduate Project Title

Initial / Zeroth / Phase-I / Final Review

Student Name 1 (Register No.)
Student Name 2 (Register No.)
Student Name 3 (Register No.)

Guided by: Dr. Guide Name, Designation

Department of Computer Science and Engineering
College / University Name

Academic Year 2026–2027

Presentation Outline

Background and Motivation

- Introduce the application domain and its significance.
- Describe the current situation using one or two key facts.
- Explain who is affected and why improvement is necessary.
- State the motivation for selecting this project.

Core Motivation

Summarize the need for the proposed project in two concise sentences.

Problem Statement

Problem

Existing systems face **[specific limitation]**, resulting in **[measurable impact]** for **[target users/stakeholders]**.

- What is the exact technical or societal problem?
- What causes the problem?
- Why are existing approaches insufficient?
- What constraints must the solution satisfy?

Project Objectives

1. To analyse **[domain/problem]** and identify its key requirements.
2. To design a **[system/model/framework]** for **[purpose]**.
3. To implement the proposed solution using **[tools/technologies]**.
4. To evaluate the solution using **[metrics/test criteria]**.
5. To compare the proposed solution with **[baseline/existing system]**.

Scope and Expected Outcomes

Included

- Target users and use cases
- Main functions/modules
- Dataset or operating environment
- Evaluation criteria

Excluded / Constraints

- Features outside the current phase
- Hardware or data limitations
- Assumptions and dependencies
- Deployment boundaries

Literature Review

Ref.	Method / Technology	Key Contribution	Limitation / Research Gap
[1]	Method A	Main finding	Limitation relevant to this project
[2]	Method B	Main finding	Limitation relevant to this project
[3]	Method C	Main finding	Limitation relevant to this project
[4]	Method D	Main finding	Limitation relevant to this project

Research Gap and Proposed Contribution

Identified Gap

- Gap 1 from prior work
- Gap 2 from prior work
- Gap 3 from prior work

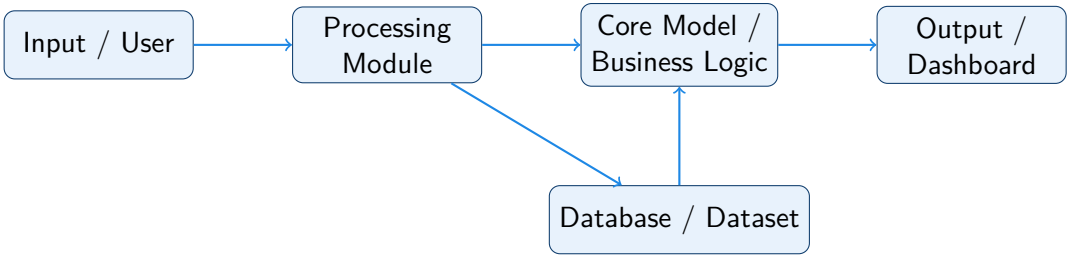
Proposed Contribution

- New feature or integration
- Technical improvement
- Expected measurable benefit

Existing System vs. Proposed System

Criterion	Existing System	Proposed System
Approach	Describe current approach	Describe proposed approach
Accuracy / Quality	Current limitation	Expected improvement
Cost / Time	Current value or issue	Expected value or benefit
Security / Privacy	Existing weakness	Proposed safeguard
Scalability	Existing constraint	Proposed capability

Proposed System Architecture



Replace the sample with the actual architecture and explain the data flow.

System Modules

1. **Module 1 – Data/Input Management:** Brief purpose and inputs.
2. **Module 2 – Preprocessing:** Cleaning, validation, or transformation.
3. **Module 3 – Core Processing:** Algorithm, model, or business logic.
4. **Module 4 – Storage/API:** Data persistence and service interfaces.
5. **Module 5 – User Interface:** Outputs, reports, and interactions.

Methodology / Workflow

1. Requirement analysis and data collection
2. Data preparation / system design
3. Model or algorithm development
4. Application integration
5. Testing and performance evaluation
6. Deployment and documentation

Evaluation Metrics

- Accuracy / F1-score
- Response time
- Resource usage
- User satisfaction

Core Algorithm / Processing Logic

Algorithm: [Insert Algorithm Name]

1. Receive and validate the input X.
2. Transform X into the required representation X'.
3. Apply the proposed method $f(\cdot)$ to obtain $Y = f(X')$.
4. Evaluate Y using the selected metric(s).
5. Return or display the final result.

Include equations, pseudocode, or a flowchart only when it improves clarity.

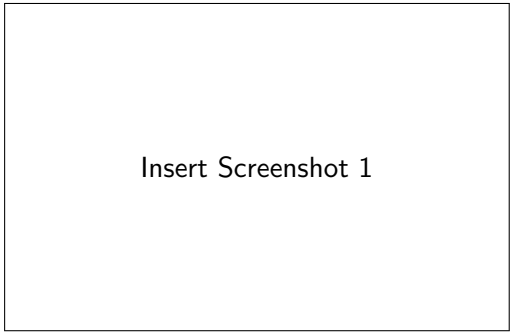
Technology Stack and Resources

Category	Selected Technology / Resource
Programming Language	Python / Java / JavaScript / Other
Framework	Django / Flask / React / TensorFlow / Other
Database	PostgreSQL / MySQL / MongoDB / Other
Dataset	Dataset name, source, size, and licence
Development Tools	Git, Docker, IDE, cloud platform, testing tools
Hardware	Processor, RAM, GPU, sensors, or embedded board

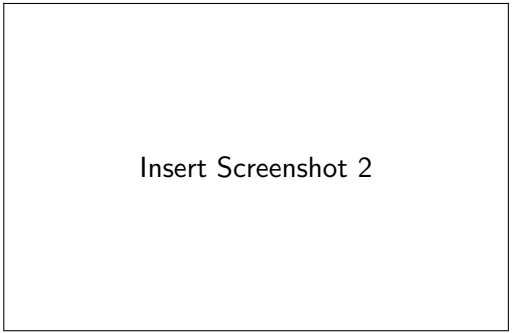
Implementation Status

Component	Status	Evidence / Remarks
Requirements and design	Completed	SRS, use cases, architecture
Data collection / setup	In progress	Mention dataset or environment
Core module	In progress	Mention completed functions
User interface	Planned	Target completion date
Testing and evaluation	Planned	Test strategy and metrics

Prototype / Implementation Screenshots



Login / Input / Main Screen



Output / Dashboard / Result Screen

Results and Performance Analysis

Insert Result Chart / Confusion Matrix

Key Findings

- Metric 1: **XX.XX%**
- Metric 2: **X.XX**
- Improvement: **XX%**
- Main observation from testing

Testing and Validation

Test ID	Test Scenario	Expected Result	Status
TC-01	Valid input / normal workflow	Correct output is generated	Pass
TC-02	Invalid or missing input	Clear validation message	Pass
TC-03	Boundary / stress condition	System remains stable	Pending
TC-04	Security / access-control check	Unauthorized access is blocked	Pass

Validation Method

Mention unit testing, integration testing, cross-validation, user testing, baseline comparison, or statistical significance tests as applicable.

Project Timeline

Activity	M1	M2	M3	M4	M5	M6
Literature review						
Requirements and design						
Implementation						
Testing and evaluation						
Documentation						

Replace M1–M6 with actual month names and update the schedule.

Team Responsibilities

Team Member	Responsibilities	Current Status
Student 1	Literature review, module design, documentation	In progress
Student 2	Data preparation, backend/model implementation	In progress
Student 3	Frontend, integration, testing and deployment	In progress

Challenges, Risks, and Mitigation

Risk / Challenge	Possible Impact	Mitigation
Limited or imbalanced data	Reduced model reliability	Augmentation and validation
Integration complexity	Schedule delay	Modular development and testing
Resource limitations	Slow execution	Optimisation or cloud resources
Security/privacy concerns	Data exposure	Access control and anonymisation

Conclusion and Future Work

Conclusion

Summarize the problem, the proposed approach, completed work, and the most important result in three or four sentences.

Future Work

- Extend the system with [feature or advanced method].
- Validate using a larger dataset or real-world deployment.
- Improve performance, security, usability, or scalability.

References I


A. Author and B. Author, "Title of the journal/conference paper," *Journal or Conference Name*, vol. X, no. Y, pp. 1–10, 2025.


C. Author, *Title of the Book*. Publisher, 2024.


Organisation Name, "Title of dataset, standard, or web resource," 2026. [Online]. Available: `https://example.org`

Thank You

Questions and Discussion

Contact: student.email@example.com